◆ TEAM 17 ◆

AI Writing vs Human Writing Survey Analytics

Executive Summary Report · April 2026

📊 DATA SCIENCE REPORT

256 Participants · 3 Analysis Phases

Python · Power BI · Scikit-Learn

256	82%	18	4
PARTICIPANTS	MODEL ACCURACY	ML FEATURES	BI DASHBOARDS

🎯 Project Overview

This project investigates how humans perceive AI-generated vs human-written text through a structured survey and a three-phase analytics pipeline, ending with a model that predicts human judgments with **82% accuracy**.

📋 Survey Design

256 participants, 21 questions covering demographics, AI opinions, and text attribution tasks (2 texts shown).

✅ Core Finding

No demographic group is significantly better at detecting AI text — AI fools everyone equally.

⚡ Strongest Signal

Text clarity is #1 predictor: very clear, well-styled text is more often attributed to AI.

🧠 Best Model

Random Forest (200 trees, max_depth=7) — 82.2% test accuracy on 224 samples.

🔄 Analytics Pipeline

Phase 1 — Descriptive Analytics (Power BI)

4 interactive dashboards: demographics, AI perception, AI vs Human comparison, and sentiment analysis.

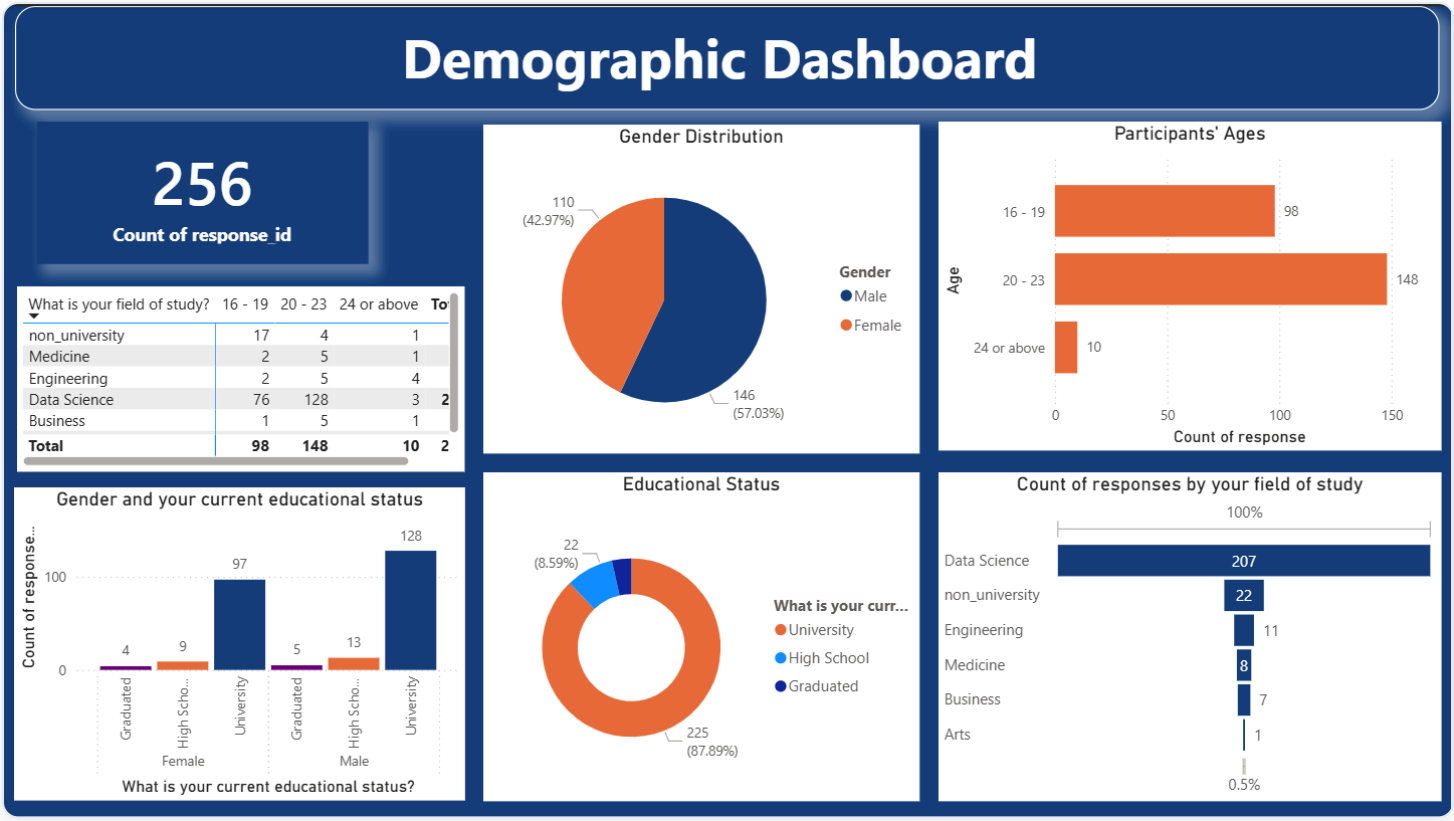
Phase 2 — Inferential Analytics (Python / scipy)

Kruskal-Wallis, Chi-Square, and Logistic Regression tests on demographic effects on AI detection ability.

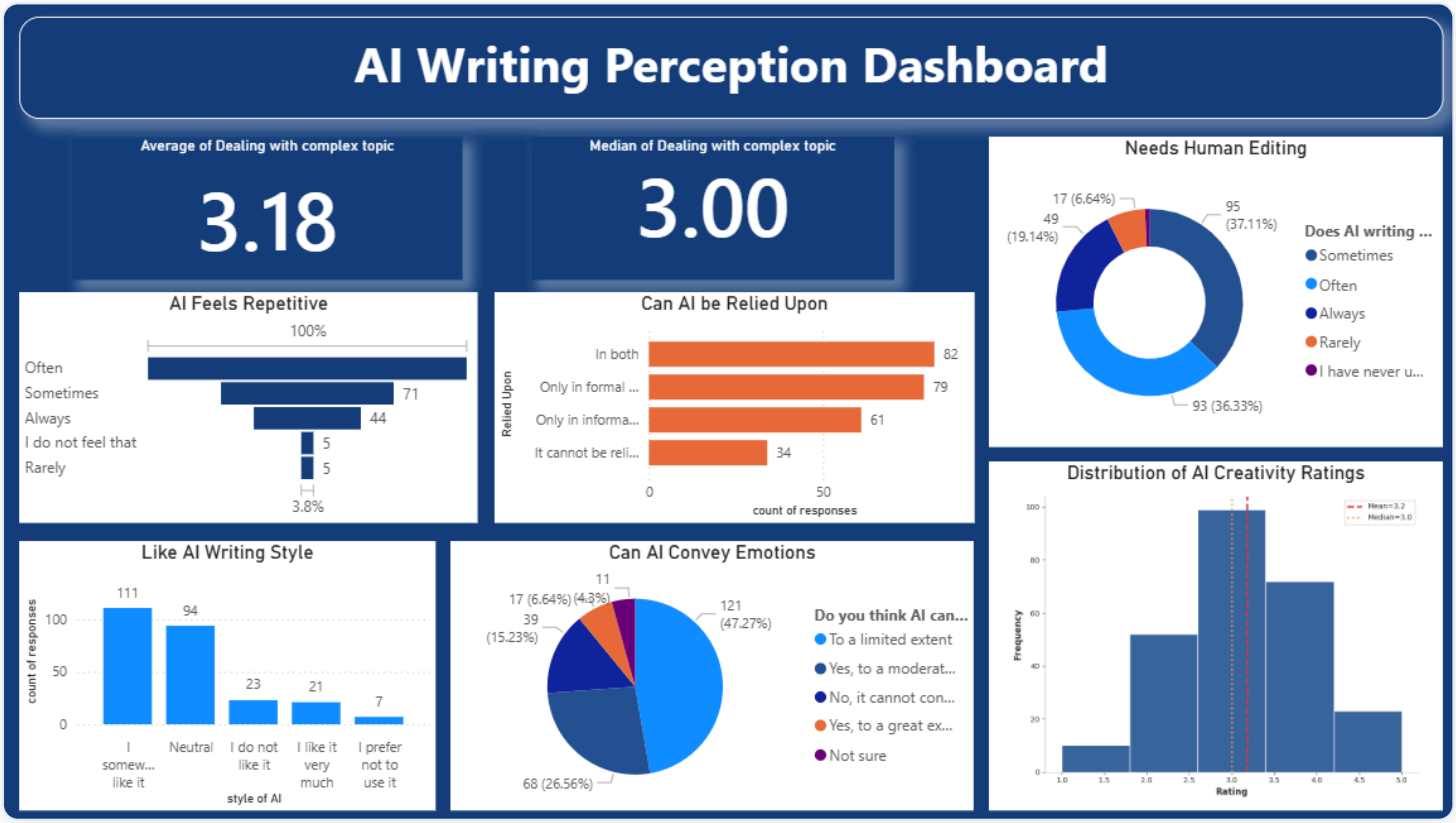
Phase 3 — Predictive Modeling (Scikit-Learn)

Random Forest predicted text attribution with 82% accuracy using 18 engineered survey features.

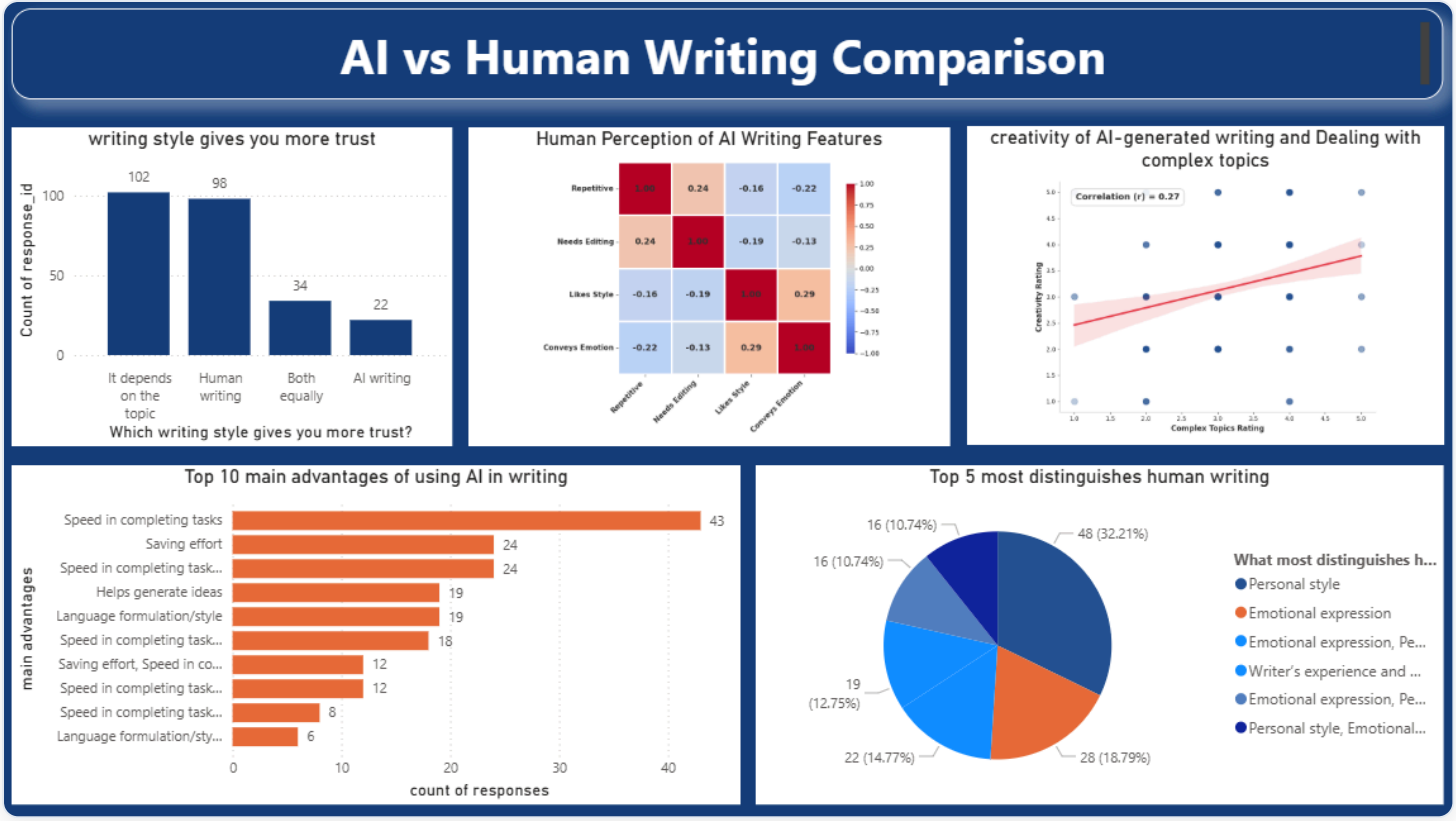
📈 Power BI Dashboards



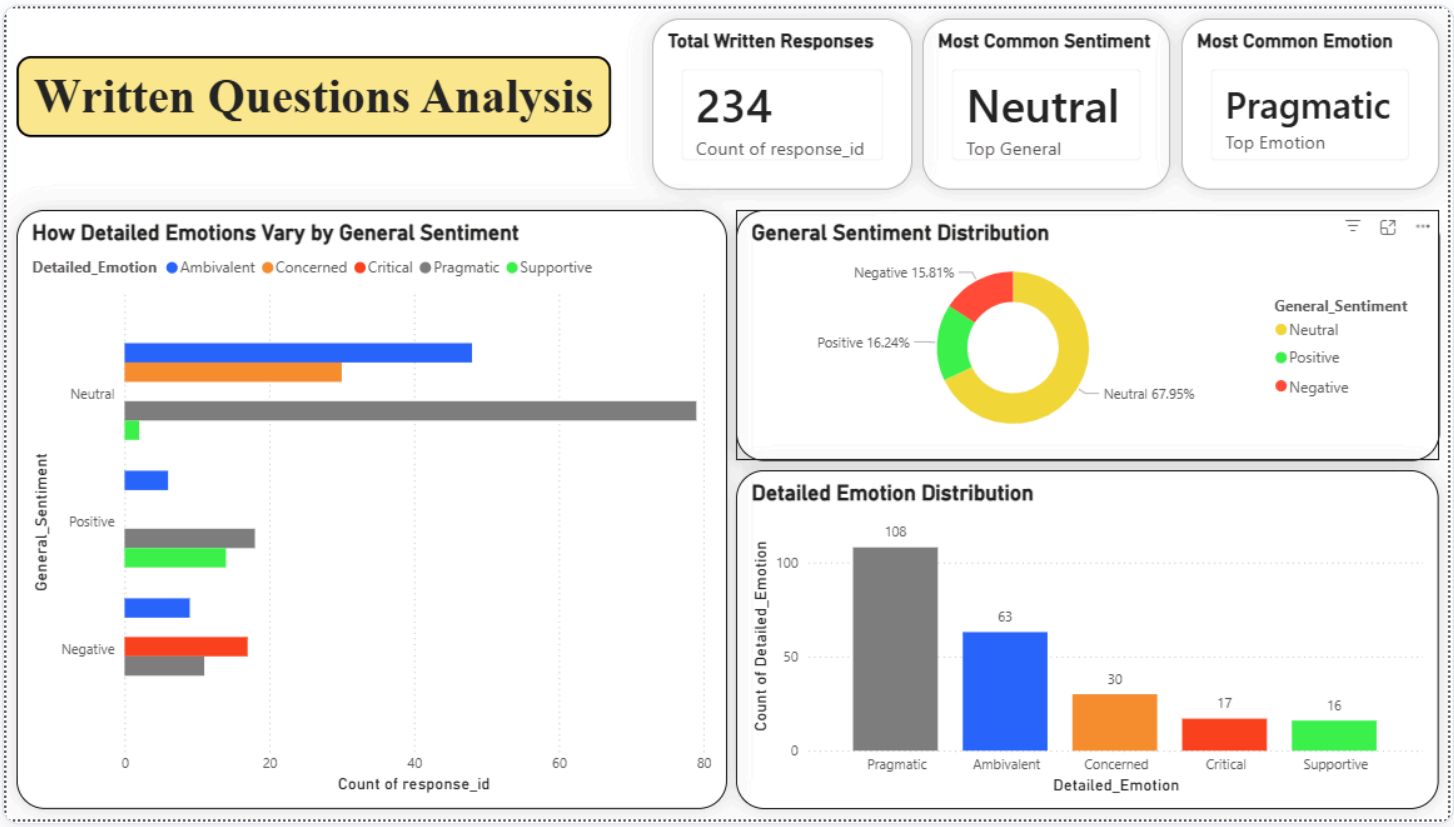
Demographic overview: mostly university students aged 16–23, predominantly in Data Science



AI writing perception ratings: creativity ~3.4/5, emotional conveyance and repetitiveness



Comparison: human writing scores higher on emotional expression; AI scores higher on clarity



Sentiment analysis of open-ended written responses by participants

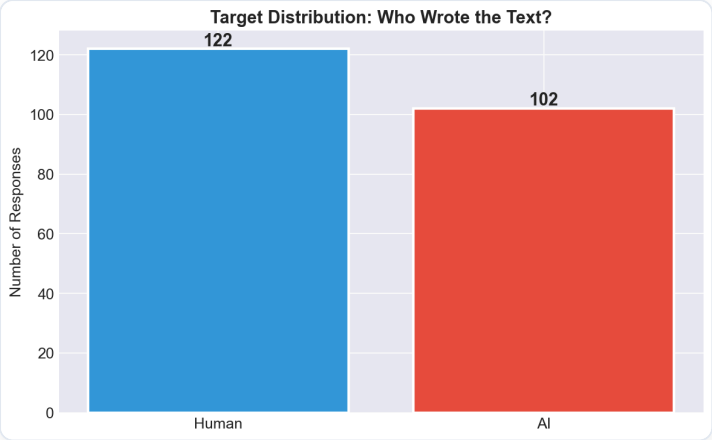
🧪 Statistical Test Results

Test	p-value	Conclusion
Age → AI Detection Score (Kruskal-Wallis)	0.146	Not significant
Field of Study → Detection Score (Kruskal-Wallis)	0.173	Not significant

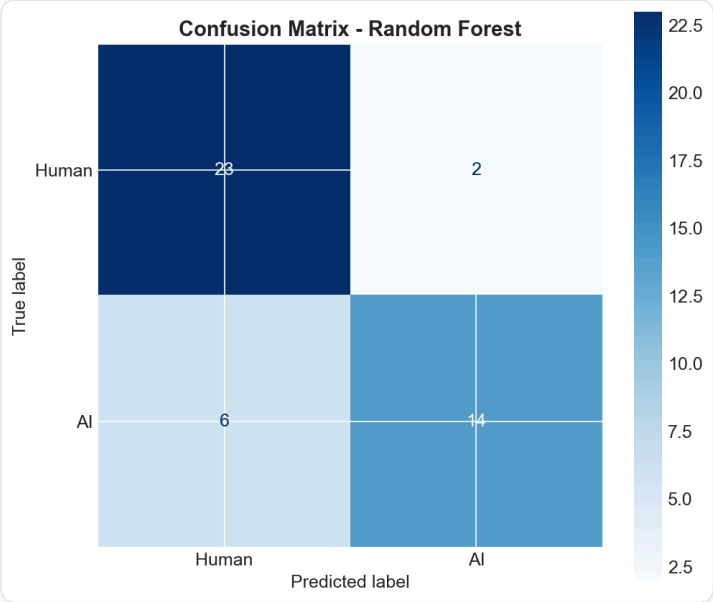
Age + Field → Good Detector (Logistic Reg.)	0.520	Weak effect
Field vs "AI Needs Editing" (Chi-Square)	0.620	No association

⚠️ **All p-values > 0.05** — demographic factors do not predict AI detection. AI fools all groups equally.

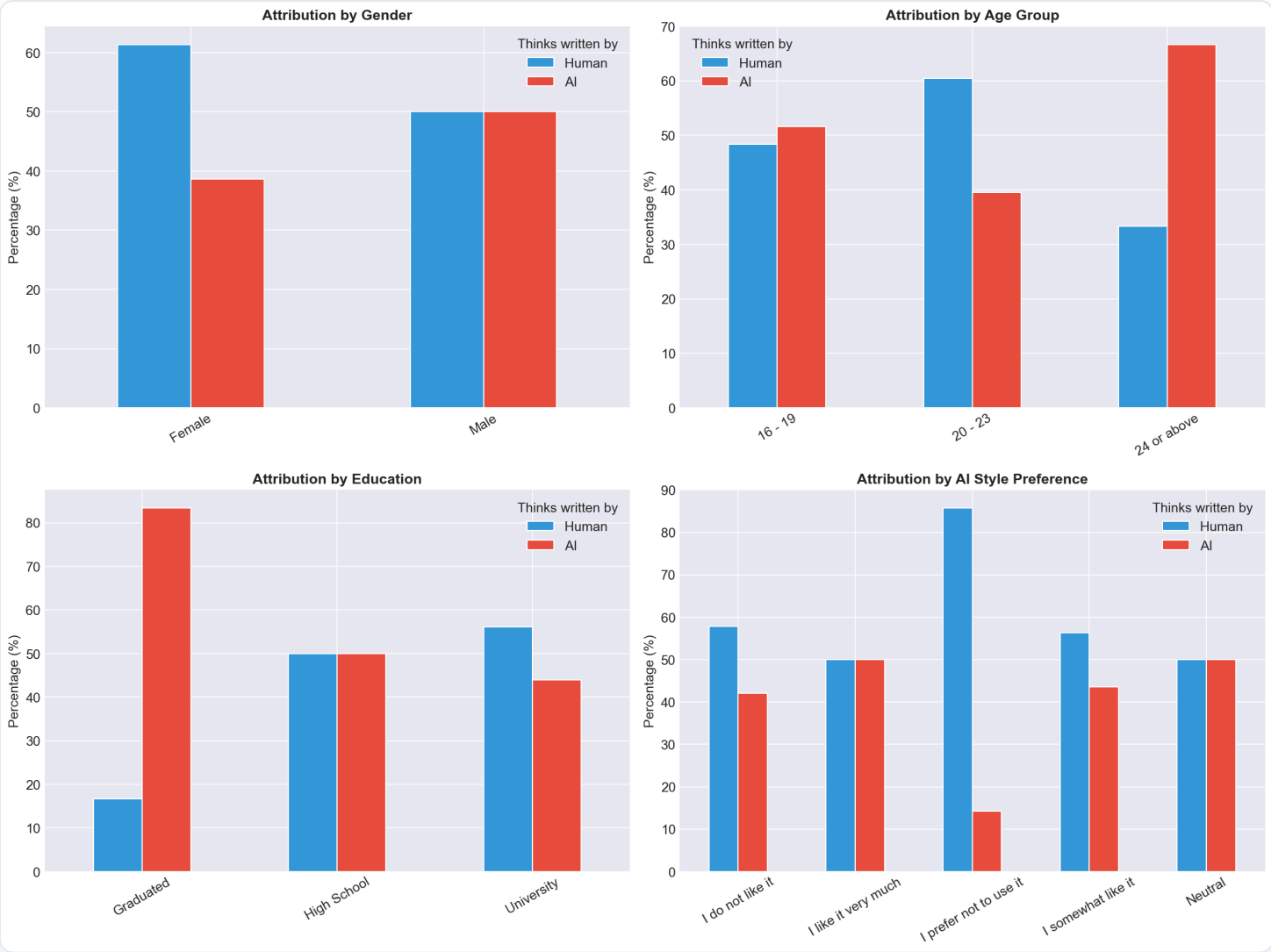
🤖 Machine Learning Results



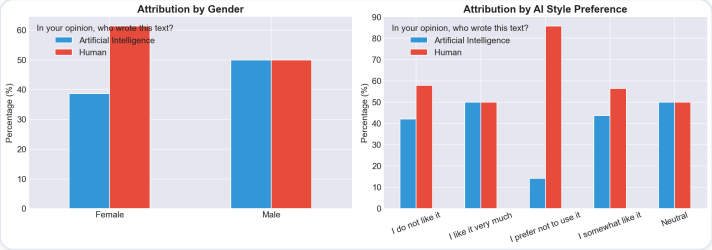
122 Human vs 102 AI — near-balanced target



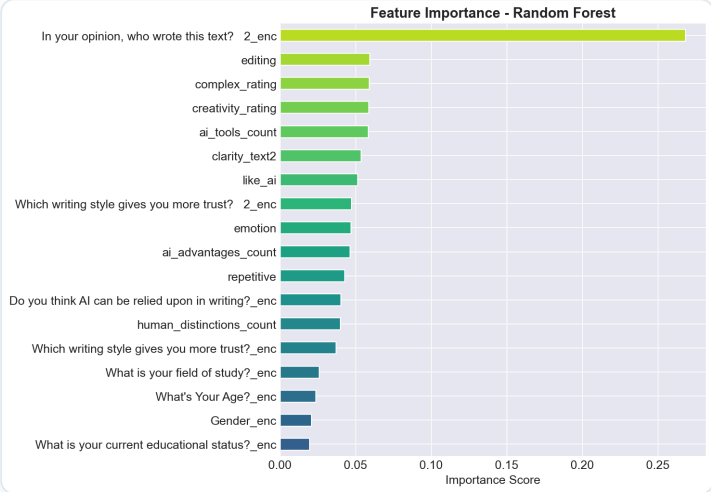
82% accuracy: 37/45 correct predictions



Attribution judgments across demographic groups — uniform pattern, no group stands out



Attribution by gender and AI preference



Top predictors: text2 attribution, editing, complexity

✅ Model Performance

Model	CV Accuracy	Test Accuracy	F1 Macro
Logistic Regression	69.25%	77.78%	0.77
Random Forest ✓	**71.52%**	**82.22%**	**0.81**

🏁 Key Takeaways

🔍 Finding 1

AI text is equally convincing across all demographics — no group detects it better than others.

📝 Finding 2

Text clarity drives attribution: clearer formatting → more often labeled as AI.

📌 Next Step 1

Expand to 1,000+ diverse participants for robust conclusions.

🚀 Next Step 2

Deploy as a web app for real-time attribution prediction.